UNIT 3 · ARS/NET PATTERN

# MCQ Bank — 175 Questions with Answers

## Calibration of seed drills, planters and plant-protection equipment (Q1–Q35)

**Q1. Calibration of a seed drill is done to:**

(a) Increase draft
(b) Determine and set the actual seed rate before field sowing
(c) Test the frame strength
(d) Measure fuel use

**Q2. Area covered by a drill of width W in N revolutions of a wheel of diameter D is:**

(a) πDNW
(b) DNW
(c) $\pi D^2NW/4$
(d) $2\pi DNW$

**Q3. A 9-row drill (20-cm spacing, W = 1.8 m) with 1.2-m wheel turned 25 revolutions covers about:**

(a) 68 m²
(b) 170 m²
(c) 340 m²
(d) 425 m²

**Q4. If 1.05 kg seed is collected over 169.6 m², the seed rate is about:**

(a) 42 kg/ha
(b) 62 kg/ha
(c) 84 kg/ha
(d) 105 kg/ha

**Q5. Revolutions needed to simulate 1/25 ha (400 m²) with D = 1.2 m, W = 1.8 m:**

(a) 25
(b) 40
(c) 59
(d) 118

**Q6. Seed rate of a fluted-roller drill is adjusted mainly by:**

(a) Tyre pressure
(b) Exposed length of the flutes
(c) Hopper size
(d) Furrow opener angle

**Q7. In an internal double-run feed, seed rate is changed chiefly by:**

(a) Drive gear/sprocket ratio
(b) Painting the rotor
(c) Seed treatment
(d) Hitch height

**Q8. The most common seed-metering device in Indian seed drills is:**

(a) Vacuum disc
(b) Fluted roller
(c) Finger pickup
(d) Belt

**Q9. Calibration must use the actual seed lot because rate depends on:**

(a) Seed size, shape, density and coating
(b) Field slope
(c) Tractor make
(d) Moon phase

**Q10. Collecting seed separately from each opener during calibration checks:**

(a) Draft
(b) Metering uniformity between openers (CV)
(c) Wheel slip
(d) Tank leaks

**Q11. Because of drive-wheel slip (5–7%), the field seed rate compared with stationary calibration is:**

(a) Higher
(b) Lower
(c) Same
(d) Double

**Q12. Seed spacing of a plate planter with wheel diameter D, cell number n and plate/wheel speed ratio K is:**

(a) $\pi D/(K \cdot n)$ (b) $K \cdot n/\pi D$

(c) $\pi DK/n$ (d) $n/(\pi DK)$

**Q13. Plant population for 75 × 20 cm spacing is:**

(a) 33,333/ha (b) 53,333/ha

(c) 66,667/ha (d) 75,000/ha

**Q14. For 75 × 25 cm spacing, population is:**

(a) 43,555/ha (b) 53,333/ha

(c) 66,667/ha (d) 80,000/ha

**Q15. In precision-planting evaluation, an interval greater than 1.5× target spacing counts in the:**

(a) Multiple index (b) Miss index

(c) Quality of feed (d) Precision

**Q16. An interval less than 0.5× target spacing indicates:**

(a) A miss (b) A double/multiple

(c) A single (d) Slip

**Q17. Quality of feed index is the percentage of intervals:**

(a) >1.5x (b) <0.5x

(c) Between 0.5x and 1.5x (d) Exactly x

**Q18. In vacuum planters, seed singulation is achieved by:**

(a) Gravity (b) Suction holding one seed per hole with a singulator

(c) Springs (d) Brushes only

**Q19. Sprayer application rate formula is Q (L/ha) =:**

(a) $600q/(v \cdot w)$ (b) $q \cdot v \cdot w/600$

(c) $60q/(v \cdot w)$ (d) $1000q/(v \cdot w)$

**Q20. Twelve nozzles of 0.8 L/min at 5 km/h with 6-m swath give:**

(a) 96 L/ha (b) 148 L/ha

(c) 192 L/ha (d) 240 L/ha

**Q21. A sprayer covers 30 m in 21.6 s. Its speed is:**

(a) 3 km/h (b) 4 km/h

(c) 5 km/h (d) 6 km/h

**Q22. Nozzle discharge varies with pressure as:**

(a) $q \propto p$ (b) $q \propto \sqrt{p}$

(c) $q \propto p^2$ (d) Independent

**Q23. To double nozzle discharge, pressure must be increased:**

(a) 2× (b) 3×

(c) 4× (d) 8×

**Q24. Large changes in application rate are best made by:**

(a) Pressure alone (b) Changing nozzle tips or forward speed

(c) Adding water (d) Raising the boom

**Q25. A nozzle should be replaced when its discharge exceeds the rated value by more than:**

(a) 2% (b) 5%

(c) 10% (d) 25%

**Q26. Increasing spray pressure makes droplets:**

(a) Coarser
(b) Finer (increasing drift risk)
(c) Unchanged
(d) Heavier

**Q27. For herbicide application the preferred nozzle is:**

(a) Hollow cone
(b) Flat fan at low pressure
(c) Solid stream
(d) Rain gun

**Q28. For insecticide/fungicide canopy penetration the common nozzle is:**

(a) Flat fan
(b) Hollow cone
(c) Flooding
(d) Anvil

**Q29. Typical boom nozzle spacing and height are:**

(a) 25 cm, 20 cm
(b) 50 cm, 45–60 cm
(c) 100 cm, 1.5 m
(d) 75 cm, 10 cm

**Q30. Knapsack sprayer calibration by the refill method measures:**

(a) Pressure
(b) Volume used over a known sprayed area
(c) Droplet size
(d) Pump wear

**Q31. Duster output for calibration is determined by:**

(a) Volume in a jar
(b) Weighing output per unit time
(c) Colour
(d) Sound

**Q32. Ultra-low-volume (ULV) spraying refers to application rates:**

(a) <5 L/ha
(b) 50–100 L/ha
(c) 200 L/ha
(d) >400 L/ha

**Q33. Tank loads needed to spray 12 ha at 200 L/ha with a 600-L tank:**

(a) 2
(b) 3
(c) 4
(d) 6

**Q34. Expected seeds dropped in N wheel revolutions from a plate with n cells and speed ratio K:**

(a) N·n·K
(b) N/(nK)
(c) πDN
(d) nK/N

**Q35. Before counting calibration output, the metering mechanism is first run a few turns to:**

(a) Warm the bearings
(b) Fill/prime the feed mechanism for steady metering
(c) Polish the rollers
(d) Check tyres

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**Answers**

Q1 — (b) Determine and set the actual seed rate before field sowing

Q2 — (a) πDNW

Q3 — (b) 170 m² — *$A = \pi DNW = 3.1416 \times 1.2 \times 25 \times 1.8 = 169.6 \approx 170\ m^2$.*

Q4 — (b) 62 kg/ha — *Rate = 10,000×1.05/169.6 = 61.9 ≈ 62 kg/ha.*

Q5 — (c) 59 — *$N = 400/(\pi \times 1.2 \times 1.8) = 400/6.786 = 58.9 \approx 59$ rev.*

Q6 — (b) Exposed length of the flutes

Q7 — (a) Drive gear/sprocket ratio

Q8 — (b) Fluted roller

Q9 — (a) Seed size, shape, density and coating

Q10 — (b) Metering uniformity between openers (CV)

Q11 — (b) Lower

Q12 — (a) πD/(K·n)

Q13 — (c) 66,667/ha — *Population = 10,000/(0.75×0.20) = 66,667.*

Q14 — (b) 53,333/ha — *Population = 10,000/(0.75×0.25) = 53,333.*
Q15 — (b) Miss index
Q16 — (b) A double/multiple
Q17 — (c) Between 0.5x and 1.5x
Q18 — (b) Suction holding one seed per hole with a singulator
Q19 — (a) 600q/(v·w)
Q20 — (c) 192 L/ha — *Q = 600×(12×0.8)/(5×6) = 600×9.6/30 = 192 L/ha.*
Q21 — (c) 5 km/h — *v = 30/21.6 = 1.389 m/s × 3.6 = 5 km/h.*
Q22 — (b) $q \propto \sqrt{p}$
Q23 — (c) 4× — *$q \propto \sqrt{p}$ → for $q_2/q_1 = 2$, $p_2/p_1 = 2^2 = 4$.*
Q24 — (b) Changing nozzle tips or forward speed
Q25 — (c) 10%
Q26 — (b) Finer (increasing drift risk)
Q27 — (b) Flat fan at low pressure
Q28 — (b) Hollow cone
Q29 — (b) 50 cm, 45–60 cm
Q30 — (b) Volume used over a known sprayed area
Q31 — (b) Weighing output per unit time
Q32 — (a) <5 L/ha
Q33 — (c) 4 — *12 ha × 200 L/ha = 2400 L ÷ 600 L = 4 tank loads.*
Q34 — (a) N·n·K
Q35 — (b) Fill/prime the feed mechanism for steady metering

## Testing and performance evaluation — tillage to sprayers and dusters (Q36–Q70)

**Q36. Farm machinery testing that uses soil bins and test rigs under controlled conditions is:**

(a) Field test
(b) Laboratory test
(c) Batch test
(d) Endurance test

**Q37. RNAM test codes were developed under:**

(a) FAO
(b) UN-ESCAP Regional Network for Agricultural Machinery
(c) WTO
(d) ISO

**Q38. The Indian Standard test code for seed-cum-fertilizer drills is:**

(a) IS 6316
(b) IS 9020
(c) IS 4468
(d) IS 12207

**Q39. Tractor test codes accepted internationally are published by:**

(a) RNAM
(b) OECD
(c) WHO
(d) BIS only

**Q40. The Central Farm Machinery Training & Testing Institute is located at:**

(a) Hisar
(b) Budni (MP)
(c) Anantapur
(d) Bhopal

**Q41. The northern region FMTTI is at:**

(a) Ludhiana
(b) Hisar
(c) Pantnagar
(d) Karnal

**Q42. Soil strength during tillage testing is characterised by:**

(a) pH meter
(b) Cone index from a penetrometer
(c) EC meter
(d) Altimeter

**Q43. Furrow cross-section and surface roughness are measured with a:**

(a) Planimeter
(b) Profilometer
(c) Pyranometer
(d) Psychrometer

**Q44. Degree of soil pulverization is expressed by:**

(a) Bulk density only
(b) Clod Mean Weight Diameter from sieve analysis
(c) Soil colour
(d) Porosity

**Q45. Clod fractions: 20% on 50 mm, 30% on 25 mm, 35% on 10 mm, 15% at mean 5 mm. MWD ≈**

(a) 12 mm
(b) 18 mm
(c) 21.8 mm
(d) 30 mm

**Q46. A finer tilth corresponds to:**

(a) Larger MWD
(b) Smaller MWD
(c) Constant MWD
(d) No MWD

**Q47. Wheel slip is computed from:**

(a) Fuel use
(b) Advance per revolution loaded vs unloaded
(c) Engine rpm only
(d) Tyre pressure

**Q48. Trash burial (inversion) evaluation applies mainly to testing of:**

(a) Sprayer
(b) Mouldboard plough
(c) Thresher
(d) Duster

**Q49. Seed-drill rate stability must be checked at different:**

(a) Hopper fill levels and machine tilts
(b) Paint colours
(c) Tyre brands
(d) Hitch pins

**Q50. Uniformity of seed spacing along the row is tested in the lab on a:**

(a) Patternator
(b) Greased/sticky belt
(c) Soil bin
(d) Test track

**Q51. Permissible visible mechanical seed damage in drill metering (BIS, cereals) is about:**

(a) 0.5%
(b) 2%
(c) 5%
(d) 10%

**Q52. Metering uniformity between rows is expressed by:**

(a) Mean only
(b) Coefficient of variation of per-opener collections
(c) Mode
(d) Range only

**Q53. Germination test of metered seed detects:**

(a) Visible damage only
(b) Internal (invisible) mechanical damage
(c) Seed colour
(d) Purity

**Q54. Field evaluation of a planter compares plant stand with:**

(a) Fuel used
(b) Target population
(c) Draft
(d) Rainfall

**Q55. Fertilizer-box materials are tested for:**

(a) Corrosion resistance
(b) Taste
(c) Transparency
(d) Magnetism

**Q56. The effective swath of a spinner spreader is fixed by:**

(a) Maximum throw distance
(b) Overlapped-pattern CV within acceptable limit (≈15%)
(c) Tank size
(d) PTO speed only

**Q57. Spreader distribution pattern is obtained using:**

(a) Collection trays across the swath
(b) Patternator grooves
(c) Penetrometer
(d) Belt

**Q58. Spray distribution across a nozzle's swath is measured on a:**

(a) Patternator
(b) Profilometer
(c) Anemometer
(d) Planimeter

**Q59. Nozzle discharge in a test should be within:**

(a) ±1%
(b) ±5%
(c) ±15%
(d) ±25% of rated

**Q60. VMD is the diameter such that:**

(a) Half the drops are smaller
(b) Half the spray VOLUME is in smaller drops
(c) All drops are equal
(d) Drift is zero

**Q61. NMD relative to VMD is always:**

(a) Larger
(b) Equal
(c) Smaller
(d) Random

**Q62. Uniformity (dispersion) ratio of a spray is:**

(a) VMD/NMD
(b) NMD/VMD
(c) VMD×NMD
(d) VMD−NMD

**Q63. A dispersion ratio approaching 1 indicates:**

(a) Very non-uniform drops
(b) Homogeneous droplet spectrum
(c) Coarse spray only
(d) High drift

**Q64. VMD 300 µm and NMD 150 µm give a uniformity ratio of:**

(a) 0.5
(b) 1.0
(c) 2.0
(d) 4.5

**Q65. Droplets are commonly sampled on:**

(a) Litmus paper
(b) Water-/oil-sensitive paper or coated slides
(c) Filter paper only
(d) Cloth

**Q66. Recommended droplet density for protectant FUNGICIDES is about:**

(a) 5–10
(b) 20–30
(c) 50–70
(d) 200–300 drops/cm²

**Q67. 'Fine' spray droplets (100–200 µm) chiefly increase:**

(a) Runoff
(b) Drift risk
(c) Draft
(d) Corrosion

**Q68. Canopy deposit distribution in field sprayer tests is assessed using:**

(a) Tracer dye/fluorescent tracer on targets
(b) Odometer
(c) Thermometer
(d) pH strips

**Q69. Duster feed rate is calibrated by:**

(a) Volume in jar
(b) Weighing output per unit time
(c) Counting particles
(d) Pressure gauge

**Q70. An endurance (durability) test primarily records:**

(a) Paint gloss

(b) Breakdowns, wear and defects over long running hours

(c) Seed rate

(d) Droplet size

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**Answers**

Q36 — (b) Laboratory test

Q37 — (b) UN-ESCAP Regional Network for Agricultural Machinery

Q38 — (a) IS 6316

Q39 — (b) OECD

Q40 — (b) Budni (MP)

Q41 — (b) Hisar

Q42 — (b) Cone index from a penetrometer

Q43 — (b) Profilometer

Q44 — (b) Clod Mean Weight Diameter from sieve analysis

Q45 — (c) 21.8 mm — *MWD = Σ(fraction × mean size) = 0.20×50 + 0.30×25 + 0.35×10 + 0.15×5 = 21.75 ≈ 21.8 mm.*

Q46 — (b) Smaller MWD

Q47 — (b) Advance per revolution loaded vs unloaded — *Slip % = (1 - distance per rev under load ÷ distance per rev unloaded) × 100; measured over a marked run counting wheel revolutions.*

Q48 — (b) Mouldboard plough

Q49 — (a) Hopper fill levels and machine tilts

Q50 — (b) Greased/sticky belt

Q51 — (a) 0.5% — *Visible damage is counted by inspection; germination testing exposes internal cracks that inspection misses (cf. combine seed damage, Unit 2 Topic 8).*

Q52 — (b) Coefficient of variation of per-opener collections

Q53 — (b) Internal (invisible) mechanical damage — *Germination testing exposes internal cracks that visual inspection misses.*

Q54 — (b) Target population

Q55 — (a) Corrosion resistance

Q56 — (b) Overlapped-pattern CV within acceptable limit (≈15%) — *Single-pass spreader patterns are deliberately overlapped; the widest bout keeping combined CV ≤ ~15% is quoted as effective swath.*

Q57 — (a) Collection trays across the swath

Q58 — (a) Patternator

Q59 — (b) ±5%

Q60 — (b) Half the spray VOLUME is in smaller drops

Q61 — (c) Smaller — *A few large drops hold most of the volume, so the number median lies well below the volume median.*

Q62 — (b) NMD/VMD

Q63 — (b) Homogeneous droplet spectrum

Q64 — (a) 0.5 — *Ratio = 150/300 = 0.5 — a typical hydraulic-nozzle value.*

Q65 — (b) Water-/oil-sensitive paper or coated slides

Q66 — (c) 50–70

Q67 — (b) Drift risk

Q68 — (a) Tracer dye/fluorescent tracer on targets

Q69 — (b) Weighing output per unit time

Q70 — (b) Breakdowns, wear and defects over long running hours

## Field capacity; testing and losses of harvesters, combines, threshers, chaff cutters; special planters (Q71–Q105)

**Q71. Theoretical field capacity (ha/h) of a machine of width W (m) at speed S (km/h) is:**

(a) WS
(b) WS/10
(c) WS/100
(d) 10WS

**Q72. A 3-m drill at 4 km/h has TFC of:**

(a) 0.75 ha/h
(b) 1.2 ha/h
(c) 12 ha/h
(d) 2.4 ha/h

**Q73. If its effective capacity is 0.9 ha/h, field efficiency is:**

(a) 60%
(b) 70%
(c) 75%
(d) 90%

**Q74. Field efficiency accounts for:**

(a) Only speed loss
(b) Turning, refilling, overlap and idle time
(c) Fuel quality
(d) Soil texture only

**Q75. A 4.2-m combine at 5 km/h with 70% field efficiency covers per hour about:**

(a) 1.05 ha
(b) 1.47 ha
(c) 2.1 ha
(d) 2.94 ha

**Q76. Time to combine 25 ha at EFC 1.47 ha/h ≈**

(a) 12 h
(b) 17 h
(c) 25 h
(d) 34 h

**Q77. The field operation typically having the LOWEST field efficiency is:**

(a) Ploughing
(b) Harrowing
(c) Boom spraying (frequent refills)
(d) Rotavation

**Q78. Capacity of threshers and combines is expressed on a:**

(a) Area basis only
(b) Material (throughput) basis — kg/h grain or kg/s MOG
(c) Fuel basis
(d) Width basis

**Q79. Pre-harvest (natural shatter) loss is measured by:**

(a) Frames placed AFTER combining
(b) Frames placed in standing crop BEFORE harvest
(c) Tank sampling
(d) Weighing straw

**Q80. Header loss equals grain found behind the header:**

(a) Plus pre-harvest loss
(b) Minus pre-harvest loss
(c) Divided by yield
(d) Times swath

**Q81. Unthreshed grain in combine discharge is charged to the:**

(a) Header
(b) Cylinder
(c) Shoe
(d) Reel

**Q82. Free grain found in the straw from the rear of a combine is the:**

(a) Shoe loss
(b) Walker/rack loss
(c) Header loss
(d) Pre-harvest loss

**Q83. 60 grains/m² of wheat (1000-grain wt 40 g) equals a loss of:**

(a) 2.4 kg/ha
(b) 12 kg/ha
(c) 24 kg/ha
(d) 60 kg/ha

**Q84. If crop yield is 4 t/ha, a 24 kg/ha machine loss is:**

(a) 0.06% (b) 0.6%

(c) 6% (d) 2.4%

**Q85. Acceptable TOTAL combine loss is commonly taken as up to about:**

(a) 0.5% (b) 3%

(c) 10% (d) 15%

**Q86. The Indian test code for power threshers is:**

(a) IS 6284 (b) IS 6316

(c) IS 4468 (d) IS 730

**Q87. Safety requirements for power threshers are covered in:**

(a) IS 9020 (b) IS 2062

(c) IS 1239 (d) IS 808

**Q88. Threshing efficiency =**

(a) 100 − % broken grain (b) 100 − % unthreshed grain

(c) Clean grain/total material (d) Grain/straw ratio

**Q89. With 10 kg unthreshed out of 1000 kg total grain, threshing efficiency is:**

(a) 90% (b) 95%

(c) 99% (d) 100%

**Q90. Cleaning efficiency is computed at the:**

(a) Straw outlet (b) Main grain outlet

(c) Feeding chute (d) Blower

**Q91. Main outlet delivers 950 kg material with 931 kg clean grain. Cleaning efficiency ≈**

(a) 93% (b) 95%

(c) 98% (d) 99%

**Q92. A thresher using 5 kW at 500 kg grain/h has specific energy:**

(a) 0.1 kWh/q (b) 1 kWh/q

(c) 10 kWh/q (d) 100 kWh/q

**Q93. Thresher capacity is reported at:**

(a) Any feed rate (b) Maximum feed rate within specified loss/breakage limits

(c) Zero feed (d) Half speed

**Q94. Grain blown out with bhusa constitutes the:**

(a) Sieve loss (b) Blowing/chaff loss

(c) Spilled loss (d) Cylinder loss

**Q95. Chaff cutter length of cut with feed velocity $v_f$, n knives, flywheel speed N is:**

(a) $v_f/(nN)$ (b) $nN/v_f$

(c) $v_f \cdot n \cdot N$ (d) $N/(n \cdot v_f)$

**Q96. $v_f$ = 15 m/min, 2 knives, 300 rpm → length of cut =**

(a) 12.5 mm (b) 25 mm

(c) 50 mm (d) 75 mm

**Q97. To get FINER chaff one should:**

(a) Reduce knives (b) Slow the flywheel

(c) Reduce feed-roller speed (or add knives) (d) Widen throat

**Q98. Chaff cutter throughput capacity is proportional to:**

(a) Throat area × bulk density × feed velocity (b) Knife colour

(c) Flywheel weight only (d) Frame length

**Q99. In rice transplanter testing, an important acceptance parameter is:**

(a) Missing hills % (≤ ~5%) (b) Seed rate kg/ha

(c) VMD (d) MWD

**Q100. 'Seedlings per hill' distribution is recorded while testing a:**

(a) Potato planter (b) Rice transplanter

(c) Sugarcane planter (d) Cotton drill

**Q101. In sugarcane cutter-planter tests, blade condition is judged by:**

(a) Bud damage % of setts (b) Paint wear

(c) Noise (d) Sett colour

**Q102. Potato planter evaluation gives special attention to:**

(a) Tuber bruising/damage and spacing (b) Droplet size

(c) Threshing (d) Winnowing

**Q103. A riding-type rice transplanter typically achieves about:**

(a) 0.05 ha/h (b) 0.3–0.4 ha/h

(c) 1.5 ha/h (d) 3 ha/h

**Q104. Grain loss in g/m² is converted to kg/ha by multiplying by:**

(a) 1 (b) 10

(c) 100 (d) 1000

**Q105. Combine grain-tank sample is analysed for:**

(a) Cleanliness and damage % (b) Draft

(c) Slip (d) MWD

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**Answers**

Q71 — (b) WS/10

Q72 — (b) 1.2 ha/h — *TFC = 3×4/10 = 1.2 ha/h.*

Q73 — (c) 75% — *η_f = 0.9/1.2 = 75%.*

Q74 — (b) Turning, refilling, overlap and idle time

Q75 — (b) 1.47 ha — *EFC = (4.2×5/10)×0.7 = 1.47 ha/h.*

Q76 — (b) 17 h — *25/1.47 = 17 h.*

Q77 — (c) Boom spraying (frequent refills)

Q78 — (b) Material (throughput) basis — kg/h grain or kg/s MOG

Q79 — (b) Frames placed in standing crop BEFORE harvest

Q80 — (b) Minus pre-harvest loss

Q81 — (b) Cylinder

Q82 — (b) Walker/rack loss

Q83 — (c) 24 kg/ha — *60 × 0.040 g = 2.4 g/m²; ×10 → 24 kg/ha.*

Q84 — (b) 0.6% — *24/4000 = 0.6%.*

Q85 — (b) 3%

Q86 — (a) IS 6284

Q87 — (a) IS 9020

Q88 — (b) 100 - % unthreshed grain

Q89 — (c) 99% — *(1000-10)/1000 = 99%.*

Q90 — (b) Main grain outlet

Q91 — (c) 98% — *931/950 = 98%.*
Q92 — (b) 1 kWh/q — *5 kW ÷ 0.5 t/h = 10 kWh/t = 1 kWh/q (1 t = 10 q).*
Q93 — (b) Maximum feed rate within specified loss/breakage limits
Q94 — (b) Blowing/chaff loss
Q95 — (a) v_f/(nN)
Q96 — (b) 25 mm — *Cuts/min = 2×300 = 600; L = 15,000 mm/600 = 25 mm.*
Q97 — (c) Reduce feed-roller speed (or add knives)
Q98 — (a) Throat area × bulk density × feed velocity
Q99 — (a) Missing hills % (≤ ~5%)
Q100 — (b) Rice transplanter
Q101 — (a) Bud damage % of setts
Q102 — (a) Tuber bruising/damage and spacing
Q103 — (b) 0.3–0.4 ha/h
Q104 — (b) 10 — *g/m² × 10 = kg/ha (1 ha = 10,000 m²; 1 kg = 1000 g).*
Q105 — (a) Cleanliness and damage %

## Selection, cost analysis, power–implement matching and reliability (Q106–Q140)

**Q106. Straight-line annual depreciation is:**
(a) C/L
(b) (C−S)/L
(c) (C+S)/L
(d) C×S/L

**Q107. Salvage value of farm machines is commonly assumed as:**
(a) 1%
(b) 10%
(c) 25%
(d) 50% of purchase price

**Q108. The economic life of an agricultural tractor is usually taken as:**
(a) 2,000 h
(b) 5,000 h
(c) 10,000 h
(d) 25,000 h

**Q109. Interest in cost analysis is charged on:**
(a) Purchase price
(b) Salvage value
(c) Average investment (C+S)/2
(d) Fuel cost

**Q110. Which is a FIXED cost?**
(a) Fuel
(b) Depreciation
(c) Repairs
(d) Wages

**Q111. Which is a VARIABLE cost?**
(a) Insurance
(b) Shelter
(c) Fuel and lubricants
(d) Interest

**Q112. Taxes, insurance and shelter together are typically about:**
(a) 0.2%
(b) 2%
(c) 8%
(d) 15% of purchase price per year

**Q113. Lubricant cost is commonly estimated as:**
(a) 5%
(b) 20–30%
(c) 50%
(d) 75% of fuel cost

**Q114. Tractor: C = ₹8,00,000, S = 10%, L = 10 yr. Straight-line depreciation is:**

(a) ₹60,000 (b) ₹72,000
(c) ₹80,000 (d) ₹88,000/yr

**Q115. Same tractor at i = 10%: annual interest =**

(a) ₹40,000 (b) ₹44,000
(c) ₹80,000 (d) ₹8,800

**Q116. With annual use, the FIXED cost per hectare:**

(a) Rises (b) Falls
(c) Constant (d) Doubles

**Q117. Break-even area for ownership vs custom hiring is:**

(a) FC×(R-VC) (b) FC/(R-VC)
(c) (R-VC)/FC (d) R/VC

**Q118. FC ₹20,000/yr, custom rate ₹800/ha, own variable cost ₹400/ha → break-even use:**

(a) 25 (b) 50
(c) 80 (d) 100 ha/yr

**Q119. Below the break-even annual use, it is cheaper to:**

(a) Buy the machine (b) Hire custom service
(c) Buy two machines (d) Idle

**Q120. In sum-of-the-years-digits depreciation, early-year depreciation is:**

(a) Lower than straight line (b) Higher than straight line
(c) Zero (d) Equal

**Q121. A key economic feature of bullock power is:**

(a) No fixed cost (b) Feed cost continues even on idle days
(c) No feed needed (d) Infinite life

**Q122. Machine replacement is economically indicated when:**

(a) Paint fades (b) Rising repair cost + obsolescence exceed value of remaining service
(c) After 1 year (d) Never

**Q123. Timeliness sizing: 40 ha in 10 days × 8 h/day needs an effective capacity of:**

(a) 0.25 (b) 0.5
(c) 1.0 (d) 2.0 ha/h

**Q124. For EFC 0.5 ha/h at η_f 75%, required TFC is about:**

(a) 0.375 (b) 0.5
(c) 0.667 (d) 1.0 ha/h

**Q125. Width to give TFC 0.667 ha/h at 4 km/h:**

(a) 1.2 m (b) 1.67 m
(c) 2.5 m (d) 3.3 m

**Q126. A 35-kW tractor with 60% power delivery gives drawbar power of:**

(a) 15 kW (b) 21 kW
(c) 28 kW (d) 35 kW

**Q127. Pull available from 21 kW at 5 km/h (1.39 m/s) is about:**

(a) 8 kN (b) 12 kN
(c) 15 kN (d) 21 kN

**Q128. With unit draft 70 kN/m² at 15 cm depth, draft per metre width is:**

(a) 1.05 kN
(b) 4.7 kN
(c) 10.5 kN
(d) 70 kN

**Q129. Hence the matched plough width for 15.1 kN pull is about:**

(a) 0.7 m
(b) 1.4 m
(c) 2.8 m
(d) 4.2 m

**Q130. For best economy a tractor should run loaded to about:**

(a) 30–40%
(b) 50%
(c) 75–85%
(d) 100–110% of rated power

**Q131. For light loads, fuel is saved by:**

(a) High gear, reduced throttle (gear-up throttle-down)
(b) Low gear, full throttle
(c) Half clutch
(d) Extra ballast

**Q132. Typical energy requirement of MB ploughing is about:**

(a) 3–5
(b) 10–15
(c) 25–40
(d) 100–150 kWh/ha

**Q133. A rotavator reduces drawbar demand because it:**

(a) Has no blades
(b) Generates forward thrust (negative draft) via PTO power
(c) Is lighter
(d) Cuts shallow only

**Q134. Reliability with constant failure rate is:**

(a) $R = 1-\lambda t$
(b) $R = e^{(-t/MTBF)}$
(c) $R = t/MTBF$
(d) $R = \lambda^2 t$

**Q135. At an operating time equal to MTBF, reliability is about:**

(a) 13.5%
(b) 36.8%
(c) 50%
(d) 63.2%

**Q136. Four subsystems in series, each R = 0.95: system reliability ≈**

(a) 0.95
(b) 0.86
(c) 0.815
(d) 0.75

**Q137. Availability =**

(a) MTTR/(MTBF+MTTR)
(b) MTBF/(MTBF+MTTR)
(c) MTBF×MTTR
(d) 1/MTBF

**Q138. MTBF 90 h and MTTR 10 h give availability of:**

(a) 80%
(b) 88%
(c) 90%
(d) 99%

**Q139. The 'infant mortality' zone of the bathtub curve shows:**

(a) Rising failures with wear
(b) High early failure rate that falls with run-in
(c) Constant failures
(d) No failures

**Q140. Most farm machines behave as which reliability system?**

(a) Parallel
(b) Series — any subsystem failure stops the machine
(c) Standby
(d) Redundant

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**Answers**

Q106 — (b) (C-S)/L

Q107 — (b) 10%
Q108 — (c) 10,000 h
Q109 — (c) Average investment (C+S)/2
Q110 — (b) Depreciation
Q111 — (c) Fuel and lubricants
Q112 — (b) 2%
Q113 — (b) 20–30%
Q114 — (b) ₹72,000 — *D = (800000-80000)/10 = 72,000.*
Q115 — (b) ₹44,000 — *Interest = (800000+80000)/2 × 0.10 = 44,000.*
Q116 — (b) Falls
Q117 — (b) FC/(R-VC)
Q118 — (b) 50 — *A_be = 20000/(800-400) = 50 ha/yr — the own-vs-hire crossover in the figure.*
Q119 — (b) Hire custom service
Q120 — (b) Higher than straight line
Q121 — (b) Feed cost continues even on idle days
Q122 — (b) Rising repair cost + obsolescence exceed value of remaining service
Q123 — (b) 0.5 — *EFC = 40/80 = 0.5 ha/h.*
Q124 — (c) 0.667 — *TFC = 0.5/0.75 = 0.667 ha/h.*
Q125 — (b) 1.67 m — *W = 10×TFC/S = 10×0.667/4 = 1.67 m.*
Q126 — (b) 21 kW — *35×0.6 = 21 kW.*
Q127 — (c) 15 kN — *Pull = 21/1.39 = 15.1 kN.*
Q128 — (c) 10.5 kN — *Per-metre draft = 70×0.15 = 10.5 kN/m.*
Q129 — (b) 1.4 m — *W = 15.1/10.5 = 1.44 m.*
Q130 — (c) 75–85%
Q131 — (a) High gear, reduced throttle (gear-up throttle-down)
Q132 — (c) 25–40
Q133 — (b) Generates forward thrust (negative draft) via PTO power
Q134 — (b) R = e^(-t/MTBF)
Q135 — (b) 36.8% — *R = e^(-1) = 0.368.*
Q136 — (c) 0.815 — *$0.95^4 = 0.8145$.*
Q137 — (b) MTBF/(MTBF+MTTR)
Q138 — (c) 90% — *90/(90+10) = 0.90.*
Q139 — (b) High early failure rate that falls with run-in
Q140 — (b) Series — any subsystem failure stops the machine

## Engine power, electric motors, batteries and standards (Q141–Q175)

**Q141. Brake power in kW from torque T (N·m) at N rpm is:**

(a) NT/60
(b) 2πNT/60,000
(c) πNT/30,000×2
(d) NT/9550 only in W

**Q142. T = 200 N·m at 2000 rpm gives BP ≈**

(a) 21 kW
(b) 33.5 kW
(c) 41.9 kW
(d) 62.8 kW

**Q143. For a 4-stroke engine, power strokes per minute equal:**

(a) rpm
(b) rpm/2
(c) 2×rpm
(d) rpm/4

**Q144. Mechanical efficiency of an engine is:**

(a) IP/BP
(b) BP/IP
(c) FP/IP
(d) BP×IP

**Q145. Friction power equals:**

(a) BP−IP
(b) IP−BP
(c) IP+BP
(d) BP/IP

**Q146. The prony brake and rope brake are examples of:**

(a) Transmission dynamometers
(b) Absorption dynamometers
(c) Chassis dynos
(d) Flow meters

**Q147. In a rope brake with dead load W, spring balance S and effective radius R, torque =**

(a) (W+S)R
(b) (W−S)R
(c) WR/S
(d) SR−W

**Q148. Tractor PTO power tests commonly use:**

(a) Rope brake
(b) Hydraulic or eddy-current dynamometer
(c) Spring balance
(d) Prony block

**Q149. A torsion (strain-gauge shaft) dynamometer:**

(a) Absorbs all power
(b) Measures torque while transmitting power
(c) Measures fuel
(d) Measures slip

**Q150. The Morse test is applicable to:**

(a) Single-cylinder engines
(b) Multi-cylinder engines
(c) Electric motors
(d) Batteries

**Q151. 4-cyl engine BP = 40 kW; cut-out readings 28, 28.5, 28, 27.5 kW. Indicated power =**

(a) 44
(b) 46
(c) 48
(d) 52 kW

**Q152. Its mechanical efficiency ≈**

(a) 75%
(b) 80%
(c) 83.3%
(d) 90%

**Q153. Willan's line method estimates friction power for:**

(a) SI engines only
(b) CI (diesel) engines at constant speed
(c) Motors
(d) Gas turbines

**Q154. In Willan's method, FP is read from:**

(a) Slope of the line
(b) Extrapolated negative intercept on the power axis at zero fuel
(c) Peak point
(d) Area under curve

**Q155. SFC 0.25 kg/kWh with CV 42,500 kJ/kg gives brake thermal efficiency ≈**

(a) 24%
(b) 30%
(c) 34%
(d) 42%

**Q156. A good tractor diesel at rated load has SFC of about:**

(a) 150
(b) 250
(c) 400
(d) 600 g/kWh

**Q157. The standard electric motor for farm use (3-φ available) is:**

(a) DC series
(b) Squirrel-cage induction
(c) Synchronous
(d) Universal

**Q158. Synchronous speed of a 4-pole, 50-Hz motor is:**

(a) 750 (b) 1000

(c) 1500 (d) 3000 rpm

**Q159. If it runs at 1440 rpm, slip =**

(a) 2% (b) 4%

(c) 6% (d) 8%

**Q160. Full-load slip of induction motors is typically:**

(a) 0% (b) 3–5%

(c) 15–20% (d) 40%

**Q161. Breakdown (pull-out) torque of an induction motor is about:**

(a) Equal to rated (b) 2–2.5× rated

(c) 10× rated (d) 0.5× rated

**Q162. For high starting torque on single-phase supply, use a:**

(a) Plain split-phase motor (b) Capacitor-start motor

(c) Shaded pole (d) Synchronous motor

**Q163. Three-phase motor input power is measured by:**

(a) One ammeter (b) Two-wattmeter method

(c) Hydrometer (d) Tachometer only

**Q164. $P = \sqrt{3} \cdot V \cdot I \cdot \cos\varphi$; $\cos\varphi$ here is the:**

(a) Slip (b) Power factor

(c) Efficiency (d) Torque ratio

**Q165. Efficiency of well-loaded induction motors is about:**

(a) 40–50% (b) 60–70%

(c) 85–92% (d) 100%

**Q166. Each cell of a lead-acid battery gives about:**

(a) 1.1 V (b) 1.5 V

(c) 2.1 V (d) 3.7 V

**Q167. Specific gravity of electrolyte in a FULLY CHARGED lead-acid battery is about:**

(a) 1.00 (b) 1.13

(c) 1.26–1.28 (d) 1.45

**Q168. Battery state of charge in the field is checked with a:**

(a) Hydrometer (b) Anemometer

(c) Manometer (d) Planimeter

**Q169. Battery capacity is rated in:**

(a) Watts (b) Ampere-hours

(c) Joules per kg only (d) Volts

**Q170. Energy stored in a 12-V, 100-Ah battery is:**

(a) 0.12 kWh (b) 1.2 kWh

(c) 12 kWh (d) 120 kWh

**Q171. Compared with lead-acid, lithium-ion batteries have:**

(a) Lower energy density (b) 4–6× higher energy density (Wh/kg) with BMS protection

(c) Same density (d) No cycle life

**Q172. The ISO technical committee for agricultural tractors and machinery is:**

(a) TC 23 (b) TC 176

(c) TC 204 (d) TC 34

**Q173. Three-point-linkage hitch categories are defined by:**

(a) ISO 500 (b) ISO 730 / IS 4468

(c) ISO 9001 (d) IS 6284

**Q174. The Indian Standards Institution (now BIS) was established in:**

(a) 1935 (b) 1947

(c) 1965 (d) 1986

**Q175. The FIRST section of an FMTTI machine test report is:**

(a) Fuel results (b) Specification verification of the machine

(c) Price list (d) Warranty

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**Answers**

Q141 — (b) 2πNT/60,000

Q142 — (c) 41.9 kW — *BP = 2π×2000×200/60,000 = 41.89 kW (or T×N/9550 = 200×2000/9550).*

Q143 — (b) rpm/2

Q144 — (b) BP/IP

Q145 — (b) IP-BP

Q146 — (b) Absorption dynamometers

Q147 — (b) (W-S)R

Q148 — (b) Hydraulic or eddy-current dynamometer

Q149 — (b) Measures torque while transmitting power

Q150 — (b) Multi-cylinder engines

Q151 — (c) 48 — *Per-cylinder IP = 40-28 = 12, 11.5, 12, 12.5 → ΣIP = 48 kW.*

Q152 — (c) 83.3% — *η_m = 40/48 = 83.3%; FP = 8 kW.*

Q153 — (b) CI (diesel) engines at constant speed

Q154 — (b) Extrapolated negative intercept on the power axis at zero fuel

Q155 — (c) 34% — *η_bt = 3600/(0.25×42500) = 33.9%.*

Q156 — (b) 250

Q157 — (b) Squirrel-cage induction

Q158 — (c) 1500 — *Ns = 120×50/4 = 1500 rpm.*

Q159 — (b) 4% — *s = (1500-1440)/1500 = 4%.*

Q160 — (b) 3–5%

Q161 — (b) 2–2.5× rated

Q162 — (b) Capacitor-start motor

Q163 — (b) Two-wattmeter method

Q164 — (b) Power factor

Q165 — (c) 85–92%

Q166 — (c) 2.1 V

Q167 — (c) 1.26–1.28

Q168 — (a) Hydrometer

Q169 — (b) Ampere-hours

Q170 — (b) 1.2 kWh — *E = V×Ah = 12×100 = 1200 Wh = 1.2 kWh.*

Q171 — (b) 4–6× higher energy density (Wh/kg) with BMS protection

Q172 — (a) TC 23

Q173 — (b) ISO 730 / IS 4468

Q174 — (b) 1947 — *ISI 1947; restructured as BIS under the BIS Act 1986 (revised Act 2016).*

Q175 — (b) Specification verification of the machine